DIGITOMICS

Student's Digital Behaviour

DETAILED REPORT

GROUP 6

Team Member	Role
Soumya Tiwari	Team Lead
Abhijeet Dey	Analysis Lead
Chirag Lalwani	Visualisation Lead
Divyanshu Singh	ETL Lead
Kshitiz	PPT and Quality Lead
Yachna	Strategy Lead

Sector: Education Technology / Student Wellbeing
Github: Github Link
Tableau Public Link: Tableau Public Link

1 Executive Summary

The Problem

Students globally spend an average of 5+ hours online every day scrolling short videos, engaging on social media, and clicking through hundreds of advertisements. School administrators, policymakers, and parents face a largely unmeasured challenge: Is this digital exposure helping or harming students? Consequences lower grades, rising anxiety, impulsive spending, reduced attention spans are felt everywhere, yet rarely backed by data at scale.

Our Approach

TheDigitomics project analysed 100,000 students across 50+ countries, spanning demographics, digital habits, academic metrics, mental health indicators, and financial behaviour. Raw data was sourced from Kaggle, cleaned using a Python ETL pipeline, and visualised through an interactive Tableau dashboard.

Key Findings

- **Late-nightdevice usage doubles anxiety.** Students who always use devices late at night score 2× higher on anxiety (6.4)thanthose who never do (3.0).
- **Everyextrasleep hour cuts stress dramatically.** Students sleeping 4 hrs face stress scores above 6; those sleeping 8+ hrsdropto~2. The relationship is linear and unambiguous.
- **Social media directly erodes productivity.** Productivity falls from ~9.5 at 0.5 hrs/day to ~7.5 at 5+ hrs/day, compounded by sleep deprivation.
- **High brain rot nearly doubles depression scores.** School-level students with high brain rot score 7.16 on depression vs.4.0 for low brain rot peers.

Key Recommendations

- Institute late-night device curfews (10 PM–6 AM) at school and hostel level — the most direct, high-impact single intervention.
- Embed structured sleep hygiene programs into school curriculum low cost, high return given the clean sleep-stress relationship.
- Target ages16–18 for cyberbullying prevention programs — exposure peaks sharply in this window before compounding into more serious harm.

> This report is written for school administrators, education policymakers, and institutional decision-makers. No statistical or technical background is required to act on these findings.

2 Sector & Business Context

The global EdTech and student wellness sector is at an inflection point. Smartphone penetration among students aged 15–25 now exceeds 80% in most countries. Platforms designed for engagement — not education — now compete directly with classrooms for student attention.

- **Attention economy vs. education:** Short-form video algorithms are engineered to maximise session length, directly competing with sustained study.
- **Mental health crisis in schools:** Depression and anxiety among students aged 15–24 have risen sharply post-2019, with digital behaviour widely cited as a contributing factor.
- **Digital inequality:** Students in underdeveloped regions face limited access to quality infrastructure and less protective digital environments.
- **Financial vulnerability:** Algorithmic advertising exposes students to hundreds of ads daily, driving impulse purchases at an age when financial literacy is lowest.

3 Problem Statement & Objectives

Formal Problem Definition: How does digital consumption behaviour i.e social media usage, short-form video, late-night device use, and ad exposure impact student academic performance, mental health, and financial decision-making? Which interventions generate the highest measurable improvement?

Success Criteria

Criterion	Status
Dataset cleaned with 0 null values	✓ Achieved
At least 8 actionable insights identified	✓ Achieved (10+ insights)
Interactive Tableau dashboard published	✓Published
Recommendations tied to quantified findings	✓ Achieved
ETL pipeline reproducible from raw data	✓ Committed to GitHub

4 Data Description

Field	Detail
Source	Kaggle **Dataset Link**
Format / License	CSV — Public / Open Access
Raw Rows → Sampled	500,000 → 100,000 (random_state=42)
Columns	48
Geographic Coverage	50+ countries
Age Range / Education Levels	15–25 years | School, Diploma, Graduate, PhD, Dropout

Key Column Groups

Group	Key Columns
Demographics	student_id, age, gender, country, urban_rural, family_income_level
Infrastructure	development_level, poverty_rate_percent, internet_infrastructure_index, avg_internet_speed_mbps
Digital Behaviour	social_media_hours, sessions_per_day, avg_session_length_minutes, late_night_usage, short_video_hours
Academic	academic_motivation, study_hours_per_week, class_attendance_rate, productivity_score
Mental Health	stress_level, anxiety_score, depression_score, sleep_hours, wellbeing_index
Financial	ads_viewed_per_day, ads_clicked_per_week, impulse_purchase_score, digital_spending_per_month
Engineered KPIs	brain_rot_index, brain_rot_level, academic_risk_score, financial_risk_score, digital_addiction_score

Data Limitations

- Cross-sectional design: Point-in-time snapshot — no causality, only correlation.
- Self-reported metrics: Stress, anxiety, depression, and motivation are self-reported and subject to response bias.
- Country-level proxies: Internet speed and poverty rate applied uniformly — ecological fallacy risk.
- Binary flags: cyberbullying_exposure and adult_content_exposure capture occurrence, not frequency or severity.

5 Data Cleaning & ETL Pipeline

All cleaning and transformation was performed in Python. Pipeline path: scripts/etl_pipeline.py

Step 1 — Extraction

- Auto-detected raw CSV (500,000 rows × 48 columns); randomly sampled 100,000 rows (random_state=42).
- Reassigned student_id sequentially (1–100,000) to restore continuity after sampling.

Step 2 — Column Standardisation

- Normalised all column names to snake_case: stripped whitespace, lowercased, replaced special characters.

Step 3 — Missing Value Treatment

- Sentinel strings ("NA", "N/A", "null", "–") replaced with NaN before imputation.
- Numeric columns: imputed with column median. Categorical columns: imputed with column mode.
- brain_rot_level : derived from brain_rot_index post-clean. Post-cleaning null count: 0.

Step 4 — Outlier Detection & Treatment

- IQR Winsorisation (factor=1.5): all numeric columns clipped to [Q1−1.5×IQR, Q3+1.5×IQR].
- Preserves distribution shape while removing extremes that distort visualisations.

Step 5 — Data Type Corrections

- Binary Yes/No columns mapped to integer flags (Yes=1, No=0). Object columns >90% parseable as numeric coerced to float64.

Step 6 — Feature Engineering

- brain_rot_level: ≤15 : Low, <=30 : Medium, >30 : High.
- academic_risk_score = (100 - attendance) × 0.4 + (10 – motivation) × 2.5 + addiction × 0.5 + brain_rot × 0.3.
- late_night_score: Never=0, Sometimes=1, Often=2, Always=3.
- financial_risk_score, digital_addiction_score, wellbeing_index: pre-computed composites validated post-clean.

Step 7 — Final Validation

- Checked 0 remaining nulls. Verified snake_case. Dropped duplicates (none found). Rounded floats to 2 d.p.
- Saved to data/processed/cleaned_data.csv — 100,000 rows × 48 columns, 0 nulls.

6 KPI & Metric Framework

	Formula / Source	Why It Matters
Brain Rot Index	Composite of short video hours, sessions/day, entertainment hours	Proxy for cognitive overload and attention degradation
Brain Rot Level	≤15=Low, ≤30=Medium, >30=High	Enables segmentation and policy targeting
Academic Risk Score	(100−att)×0.4 + (10−mot)×2.5 + addiction×0.5 + brain_rot×0.3	Combines four risk dimensions into one actionable number
Digital Addiction Score	Pre-computed composite (sessions, hours, engagement)	Leading indicator of mental health and productivity decline
Wellbeing Index	Pre-computed composite (sleep, stress, anxiety, depression)	Single-number health summary for segmentation
Financial Risk Score	Composite of ad exposure, impulse score, digital spending	Flags students at risk of financial harm
Late Night Score	Never=0, Sometimes=1, Often=2, Always=3	Enables regression and correlation analysis of sleep disruption

KPI-to-Objective Mapping

Objective	Primary KPI(s)
Measure digital impact on academics	Academic Risk Score, Brain Rot Index, Productivity Score
Understand mental health outcomes	Wellbeing Index, Anxiety Score, Depression Score, Sleep Hours
Identify financial risk	Financial Risk Score, Impulse Purchase Score, Digital Spending
Guide policy interventions	Late Night Score, Digital Addiction Score, Brain Rot Level

7 Exploratory Data Analysis

7.1 Academic Performance & Digital Habits

Brain Rot Index	Attention Span	Academic Risk Score	Internet Access	Sleep Hours
19.02	53.68 min	27.99	5.01 hrs/day	5.99 hrs

Social Media vs. Productivity

Productivity scores fall from ~9.5 at 0.5 hrs/day to ~7.5 at 5+ hrs/day. Students with fewer sleep hours cluster at the high-usage, low-productivity end — revealing a compounding cycle: more social media → less sleep → even lower productivity.

Implication: A student spending 5 hrs/day on social media is 21% less productive than one spending under 1 hr. Across a school of 1,000 students this represents a massive aggregate academic output loss.

Depression vs. Brain Rot Index

The relationship is stark and consistent across education levels:

Brain Rot Level	School Depression Score	PhD Depression Score
Low	3.998	~3.8
High	7.159	6.825

Implication: Brain rot is a clinical-scale predictor of depression severity. Schools with high smartphone use should treat this as a mental health risk factor, not just an academic one.

Infrastructure & Access Gap

Region	Student Count
Developing Region Underdeveloped	23,024 urban / 18,912 rural
Region	9,935 urban / severely limited rural access

7.2 Mental Health & Safety

Avg Wellbeing Index	Cyberbullying Exposure	Avg Stress Level	Digital Addiction	Financial Risk
57.28	12.54%	4.849	14.08	41.61

Late-Night Usage vs. Anxiety (Dose-Response)

Late-Night Usage Frequency	Avg Anxiety Score
Never	~3.0
Sometimes	~4.2
Often	~5.4
Always	~6.4

Implication: This is the clearest, most actionable finding in the dataset. The "Always" group scores 2× higher than the "Never" group. A device curfew policy has a directly measurable expected anxiety reduction.

Sleep Hours vs. Stress Level

The relationshipis linear andunambiguous. Students averaging 4 hrs of sleep face stress scores above 6; those averaging 8+ hrs drop to ~2. Every additional hour of sleep corresponds to roughly 1 point of stress reduction ($\beta \approx -1.0$, $R^2 \approx 0.42$).

Implication: Sleep is the cheapest mental health intervention available. A structured sleep hygiene program costs near-zero to implement and has a quantifiable, linear return.

Cyberbullying Exposure by Age

Exposurepeaksbetween ages16–18 (1,247–1,260 students per bracket), dips at 20, then rises again at 22–23 — reflecting high-school vulnerability and early college re-exposure. This provides a precise targeting window for prevention programs.

7.3 Spending, Addiction & Behavioural Patterns

Adult Content Exposed	Avg Ads/Day	Avg Digital Spend/Month	Posts/Week	Impulse Score
18%	97	$59.63	4.19	5.48

Urban vs. Rural Digital Addiction by Income

Segment	Avg Digital Addiction Score
Urban High Income	15.6
Urban Middle Income	14.5
Urban Low Income	13.2
Rural High Income	14.1
Rural Middle Income	13.5
Rural Low Income	12.9

Implication: Urban high-income students have maximum access, maximum spending power, and the least friction. They are the primary target for financial literacy and addiction awareness programs.

Internet Speed vs. Brain Rot Index

Counter-intuitively, faster internetcorrelates with higher brain rot — Switzerland (237.9 Mbps), France (234.3 Mbps), and Singapore (217.4 Mbps) are among the highest brain rot nations. Speed enables more passive content consumption, not better learning. Infrastructure investment alone is insufficient without digital literacy policy.

Impulse Buying by Income Level

High-incomestudents'digitalspending rises sharply beyond an impulse score of 5, while low-income students plateau. Income amplifies impulsive digital behaviour — not just a matter of access, but unchecked spending power. Financial literacy programs should be income-segmented.

8 Statistical Analysis

8.1 Correlation Analysis

Variable Pair	Direction	Strength
Social Media Hours ↔ Productivity Score	Negative	Moderate–Strong
Sleep Hours ↔ Stress Level	Negative	Strong
Late Night Score ↔ Anxiety Score	Positive	Strong
Brain Rot Index ↔ Depression Score	Positive	Moderate–Strong
Digital Addiction Score ↔ Wellbeing Index	Negative	Moderate
Impulse Purchase Score ↔ Digital Spending	Positive	Strong Weak–
Internet Speed ↔ Brain Rot Index	Positive	Moderate

8.2 Hypothesis Testing

H1:Late-night user shave significantly higher anxiety than non-users

- **Test:** t-test (Always vs Never groups)
- **Result:**$p<0.001$—Reject null. Late-night usage is a significant anxiety predictor.

H2: Higher brain rot index is associated with higher depression

- **Test:** Pearson correlation+ ANOVA across Brain Rot Level groups
- **Result:** $p < 0.001$, $r \approx 0.48$. Confirmed moderate-strong positive relationship.

H3:Sleepdurationsignificantlypredicts stress level

- **Test:** Linear regression (sleep hours → stress)
- **Result:** $\beta \approx -1.0$peradditionalsleephour,$R^2 \approx 0.42$. Strong predictor confirmed.

8.3 Segmentation & Risk Detection

Students were segmented across three axes: Brain Rot Level (Low/Medium/High), Income Level (Low/Middle/High), and Urban vs. Rural. Key finding: high brain rot + urban + high income is the highest-risk cluster for both mental health decline and financial exposure.

Academic Risk Score Tiers

Risk Tier	Academic Risk Score	% of Students
Low Risk	< 20	~25%
Moderate Risk	20–35	~50%
High Risk	> 35	~25%

Half the student population sits in the moderate-risk zone — the primary target for preventive intervention.

9 Tableau Dashboard Design

Dashboard URL: Tableau Public Link

The Tableau dashboard answers one key question for decision-makers: **Where should I intervene first?** It translates 100,000 rows of data into visual narratives requiring no statistical background to interpret.

View Structure & Interactive Filters

View	Purpose
Dashboard Page 1 — Academic Performance & Digital Habits	Executive overview of digital usage impact on academics and brain rot
Dashboard Page 2 — Mental Health & Safety	Wellbeing, anxiety, stress, and exposure risk breakdown
Dashboard Page 3 — Spending, Addiction & Behavioural Patterns	Financial risk, advertising impact, and addiction segmentation

Interactive Filters Available

Development Level · Education Level · Gender · Urban/Rural toggle · Age range slider (15–25) · Income Level

Key Views Explained

Chart	Key Insight
Scatter: Social Media Hours vs. Productivity	Negative relationship with sleep hours encoded as dot colour. Bottom-right cluster (high usage, low productivity, low sleep) is the primary intervention target.
Heatmap: Brain Rot Level × Education → Depression	3×6 matrix revealing how depression severity scales with passive digital consumption. Top-right cell (High brain rot, School) is highest-risk intersection.
Bar: Late-Night Usage → Anxiety Score	Four bars (Never/Sometimes/Often/Always) showing monotonic rise. Decision-maker sees the answer immediately: eliminate "Always" late-night usage.
Line: Sleep Hours → Stress Level	Cleanest finding — near-linear descent in stress as sleep increases. Ideal for school health campaigns.
Map: Country-Level Digital Spending & Brain Rot	Russia and Australia: highest digital spenders ($70–75/month). High-speed nations (Switzerland, France, Singapore) show elevated brain rot.
Stacked Bar: Urban/Rural × Income → Digital Addiction	Urban high-income segment is highest-addiction group — guides where targeted digital literacy investment is most needed.

10 Insights Summary

1	**Late-night device use doubles anxiety.** Students who always use devices late at night score 6.4 on anxiety vs. 3.0 for those who never do. Schools and hostels should treat device curfews as a mental health intervention, not a discipline measure.
2	**Every hour of lost sleep adds ~1 point of stress.** The sleep-stress relationship is linear, clean, and strong ($R^2 \approx 0.42$). Institutions that invest in sleep hygiene programs are directly investing in stress reduction.
3	**Social media is a productivity tax.** A student spending 5+ hrs/day on social media is ~21% less productive than one spending under 1 hr — an enormous aggregate loss addressable through policy.
4	**High brain rot nearly doubles depression scores at school level.** School-level students with a high brain rot index score 7.16 on depression vs. 3.99 for low brain rot peers. Restricting passive content is a mental health imperative.
5	**Ages 16–18 are the peak window for cyberbullying exposure.** With 1,247–1,260 students exposed per bracket, targeted prevention programs should be deployed for this cohort before harm compounds.
6	**Faster internet does not mean safer or smarter digital behaviour.** Switzerland, France, and Singapore lead on internet speed — and show elevated brain rot indices. Infrastructure must be paired with digital literacy policy.
7	**Urban high-income students carry the highest digital addiction scores (15.6).** Maximum access + spending power + least friction. They are the primary target for financial literacy and addiction awareness programs.
8	**Income amplifies impulsive digital spending.** High-income students' digital spending rises sharply above impulse score 5, while low-income students plateau. Unchecked digital commerce is a financial risk for high-income youth.
9	**Students viewing 97 ads/day are primed for habitual impulse spending.** Even a 10% redirection of that exposure toward educational content would materially shift the learning vs. passive consumption balance.
10	**Cyberbullying exposure peaks again at ages 22–23.** This secondary peak — likely linked to early career social media use — is overlooked by school-focused programs. Colleges need their own intervention layer.

11 Recommendations

Priority 1: Late-Night Device Curfews (10 PM–6 AM)

Insight	Students who always use devices late at night score 2× higher on anxiety.
Recommendation	Implement school/hostel-level policy blocking personal device internet access from 10 PM–6 AM.
Expected Impact	Projected 30–40% reduction in "Always" late-night usage; anxiety reduction of 1–2 points on average; downstream sleep and stress improvement.
Feasibility	Network-level controls are technically straightforward. Pilot in one hostel and measure anxiety scores at 6 months.

Priority 2: Embed Sleep Hygiene Into School Curriculum

Insight	Every additional hour of sleep reduces stress by ~1 point. Students at 4 hrs face stress >6; those at 8+ drop to ~2.
Recommendation	Introduce evidence-based sleep hygiene module into health education (ages 13–18), with sleep tracking in student wellness check-ins.
Expected Impact	If average sleep increases by 1 hr (5.99→6.99 hrs), projected ~1 point stress reduction. Wellbeing index improvement of 3–5 points.
Feasibility	Low cost — deliverable as a 4-week module within existing health education slots. Evidence-based content is publicly available.

Priority 3: Age-Targeted Cyberbullying Prevention (Ages 16–18)

Insight	Cyberbullying exposure peaks sharply at ages 16–18 (1,247–1,260 students per bracket) and rises again at 22–23.
Recommendation	Deploy dedicated cyberbullying awareness and reporting program targeting Grades 10–12 with anonymous reporting, peer-support counsellors, and platform literacy training.
Expected Impact	Targeting the highest-exposure cohort before harm compounds is the most cost-efficient prevention strategy. Expected 10–15% reduction in unreported incidents within one academic year.
Feasibility	Evidence-based frameworks (e.g., Cyberbullying Research Center) are freely available. Requires training of 2–3 counsellors per school.

Priority 4: Digital Financial Literacy for High-Income Urban Students

Insight	Urban high-income students show the highest digital addiction scores (15.6) and steepest impulse-buying curves above impulse score 5.
Recommendation	Introduce digital financial literacy program covering ad literacy, impulse purchase recognition, and monthly spend auditing. Target urban private schools and colleges.
Expected Impact	Potential 15–20% reduction in average digital spending ($59.63 → ~$50) in high-income cohort.
Feasibility	6-session workshop. Partnership with FinTech companies (who have reputational interest in responsible use) could fund delivery.

Priority 5: Redirect Ad Exposure Toward Educational Content

Insight	Students view an average of 97 ads per day. Faster internet enables more passive content consumption, not more learning.
Recommendation	Implement content governance on school-network devices. Use DNS-level ad-filtering to redirect 10–15% of ad exposure toward curated educational content.
Expected Impact	Shifting even 10 of 97 daily ads to educational prompts meaningfully increases the learning vs. passive consumption ratio.
Feasibility	DNS-level ad filtering on school networks is low-cost and technically simple.

12 Impact Estimation

Recommendation	Estimated Impact	Timeframe
Late-night device curfew	30–40% reduction in "Always" late-night users; anxiety drops 1–2 pts	3–6 months
Sleep hygiene curriculum	+1 hr avg sleep → –1 pt stress → +3–5 pts wellbeing index	1 academic year
Cyberbullying prevention (16–18)	10–15% reduction in incident rate among peak-exposure cohort	1 academic year
Digital financial literacy	15–20% reduction in avg monthly digital spending in high-income cohort	6–12 months
Ad redirection policy	Measurable shift in education vs. entertainment content ratio within school hours	1–3 months

Why Act Now? Digital addiction scores, brain rot indices, and anxiety levels are driven by platforms actively optimised to increase engagement year-over-year. Delay is not neutral — it compounds the harm. The interventions proposed are low-cost, high-return, and evidence-backed. Every semester of inaction is a semester of measurable student outcome degradation.

13 Limitations & Future Scope

Key Limitations

- **Cross-sectional data:** No longitudinal tracking — causality cannot be confirmed, only correlation.
- **Self-reported metrics:** Stress, anxiety, depression, motivation, and sleep are self-reported — subject to social desirability bias.
- **Country-level proxies:** Internet speed and poverty rate applied uniformly to all students from that country — ecological fallacy risk.
- **Causality cannot be concluded:** Reverse causation is possible (e.g., less productive students use more social media as an escape).
- **Individual predictions:** Aggregate patterns do not predict individual outcomes. A high brain rot index student may perform excellently.

Future Scope

Area	Description
Longitudinal Study Design	Follow the same cohort for 4–6 years to confirm whether high brain rot genuinely causes lower outcomes or is driven by a confounding factor.
Causal Inference Modelling	Propensity score matching or instrumental variable analysis to quantify effect sizes for policy cost-benefit analysis.
AI Tool Usage Layer	Track how LLM use (ChatGPT, Claude, Gemini) interacts with brain rot index, attention span, and academic risk in a new cohort study.
Policy RCTs	Randomly assign matched schools to interventions; measure outcomes at 6 and 12 months for evidence-grade support.
Gender-Disaggregated Analysis	Produce gender-stratified dashboards — current analysis does not break down outcomes by gender, a significant gap.
Real-Time Dashboard Integration	Live API connection to anonymised platform usage data powering a real-time wellbeing dashboard for school administrators.

14 Conclusion

Students today are navigating a digital environment that was not designed with their wellbeing in mind. This project quantified — across 100,000 students globally — the measurable costs of that environment: a near-doubling of depression scores for high brain rot students, a 2× anxiety gap between heavy and zero late-night users, and a linear erosion of productivity as social media hours grow.

The approach was systematic: raw data from Kaggle was cleaned through a reproducible Python ETL pipeline, engineered into actionable KPIs, and visualised in an interactive Tableau dashboard accessible to any decision-maker without statistical training.

The most important finding is also the most actionable: late-night device use is the single highest-leverage point for institutional intervention. A school or hostel that implements an evening device curfew does not need to solve the entire digital attention economy — it just needs to protect the 8 hours of sleep that its students are currently trading away for a social media feed. The data is clear. The interventions are known. The cost of acting is low. The cost of waiting is another academic year of compounding harm.

A Appendix

A. Data Dictionary

Full column definitions (all 48 columns with descriptions and value types): docs/data_dictionary.md

B. ETL Pipeline & Notebook Reference

Artifact	Path
Full ETL pipeline	scripts/etl_pipeline.py
Extraction notebook	notebooks/01_extraction.ipynb
Cleaning notebook	notebooks/02_cleaning.ipynb
EDA notebook	notebooks/03_eda.ipynb
Statistical analysis	notebooks/04_statistical_analysis.ipynb
Load/validation notebook	notebooks/05_final_load_prep.ipynb

C. Key Formulas

Academic Risk Score: (100 – attendance) × 0.4 + (10 - motivation) × 2.5 + digital_addiction × 0.5 + brain_rot_index × 0.3 [clipped to min 0]
Brain Rot Level: brain_rot_index ≤15 → Low | ≤30 → Medium | >30 → High

D. Contribution Matrix

Team Member	Dataset& Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT & Viva
Soumya Tiwari			✓	✓	✓	✓	
Abhijeet Dey	✓		✓				
Chirag Lalwani				✓	✓		
Divyanshu Singh		✓	✓				
Kshitiz						✓	✓
Yachna	✓						